Connect forces to velocity and position

LEARNING TARGET I can connect forces, velocity change, and final position.

A · KEY VOCABULARY

Chain	Net force → a → velocity and position.
Interval	Keep one start and end time throughout.
Reversal	v can cross zero while a keeps its sign.
Area	Add positive and negative pieces of Δx.

B · KEY EQUATIONS + UNITS

$$a = \Sigma F \div m$$

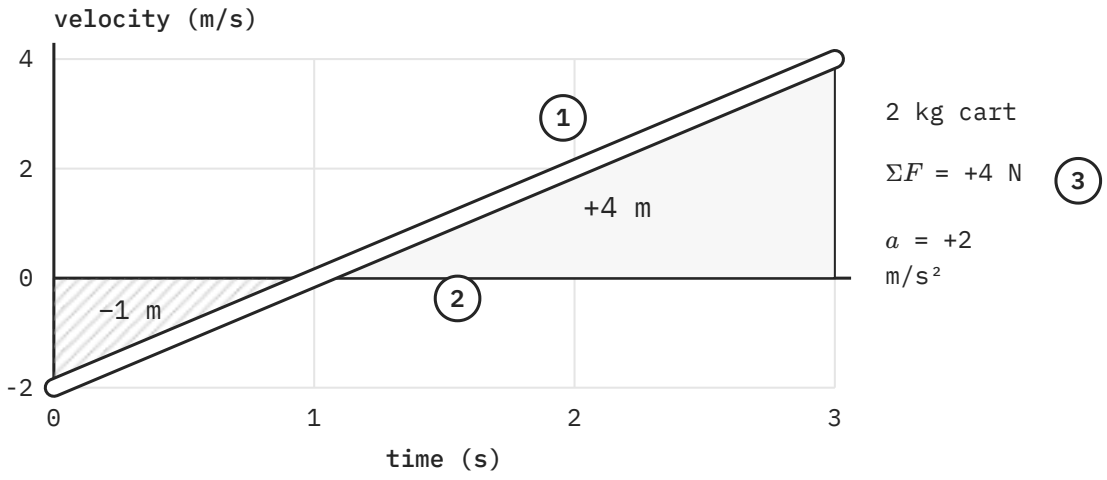
$$v_f = v_i + at$$

x_f $x_0 + v_i t + \frac{1}{2}at^2$

`model` constant a

C · COLOR IT · FORCE TO POSITION

*Velocity crosses zero at **1 s**.*



☐ **1 · PURPLE** Velocity line

☐ **2 · BLUE** Signed areas

☐ **3 · ORANGE** Net force: **+4 N**

D · WORKED EXAMPLE

A **2 kg** cart starts at **1 m** with velocity **−2 m/s**. A constant **+4 N** net force acts for **3 s**. **Find final velocity and position.**

GIVEN	EQUATION	WORK	ANSWER
m = **2 kg**; ΣF = **+4 N** x_0 = **1 m**; v_i = **−2 m/s** t = **3 s**	Find a from force. Then steady-a equations.	$a = 4/2 =$ **+2 m/s²** $v_f = -2 + 2(3)$ $x_f = 1 - 6 + 9$	**+4 m/s; +4 m** Δx = **+3 m**; distance = **5 m**.

E · THE 2026-XJ THREAD

DAY 20 OF 22 · NEW TOOL: THE FULL CHAIN

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Follow the chain from force to position

Constant net force, velocity reversal, and signed displacement

We now have the pieces needed to predict motion from forces. The net force gives acceleration. Acceleration changes velocity. Velocity over time changes position. I want each link to use the same object, reference frame, and time interval. That consistency matters especially when an object reverses direction during the calculation.

Start with the given values in **Section D**. Our **2 kg cart** starts at **1 m** with velocity **−2 m/s**. A constant net force of **+4 N** acts for **3 s**. Take right as positive. The first step is $a = \Sigma F \div m = 4\ \text{N} \div 2\ \text{kg} = +2\ \text{m/s}^2$. Because force and mass stay constant, acceleration stays constant over the interval.

Next find final velocity: $v_f = v_i + at = -2\ \text{m/s} + (2\ \text{m/s}^2)(3\ \text{s}) = +4\ \text{m/s}$. The cart began moving left and finishes moving right. Its velocity passes through zero at $t = (0 - (-2))\ \text{m/s} \div (2\ \text{m/s}^2) = 1\ \text{s}$. Positive acceleration first slows the negative motion, then builds positive motion.

The displacement equation covers the entire constant-acceleration interval, including the turn: $\Delta x = v_i t + \tfrac{1}{2}at^2$. Substitution gives $\Delta x = (-2\ \text{m/s})(3\ \text{s}) + \tfrac{1}{2}(2\ \text{m/s}^2)(3\ \text{s})^2 = -6\ \text{m} + 9\ \text{m} = +3\ \text{m}$. Finally add the starting coordinate: $x_f = 1\ \text{m} + 3\ \text{m} = 4\ \text{m}$. Displacement and final position are different quantities.

Compare the areas below and above the time axis in **Section C**. Their signed sum explains why displacement is only 3 m. During the first second, the velocity line lies below zero. Its triangular area is $-\tfrac{1}{2}(1\ \text{s})(2\ \text{m/s}) = -1\ \text{m}$. During the next two seconds, the line lies above zero, giving $+\tfrac{1}{2}(2\ \text{s})(4\ \text{m/s}) = +4\ \text{m}$. Adding the signed areas gives the same +3 m displacement. The blue regions track position change; the purple line tracks velocity.

Distance uses the sizes of those two regions instead: **distance = 1 m + 4 m = 5 m**. The cart travels from its starting coordinate of 1 m leftward to 0 m, then rightward to 4 m. That route makes the distinction concrete. A positive final displacement does not mean every part of the trip was in the positive direction.

We can check the chain in several ways. The net force and acceleration point the same way. Velocity changes by 6 m/s over three seconds at 2 m/s². The graph's signed areas agree with the displacement equation. Adding the initial position gives the correct final coordinate. If the force changed during the trip, this single constant-acceleration calculation would need to be divided into suitable intervals or replaced with a more detailed model.

For our fictional asteroid, 2026-XJ, a forecast needs both its present motion and the forces changing that motion. Starting with force alone leaves out initial velocity and position. Connecting all three ideas gives us a prediction we can explain, check, and revise when new evidence arrives.